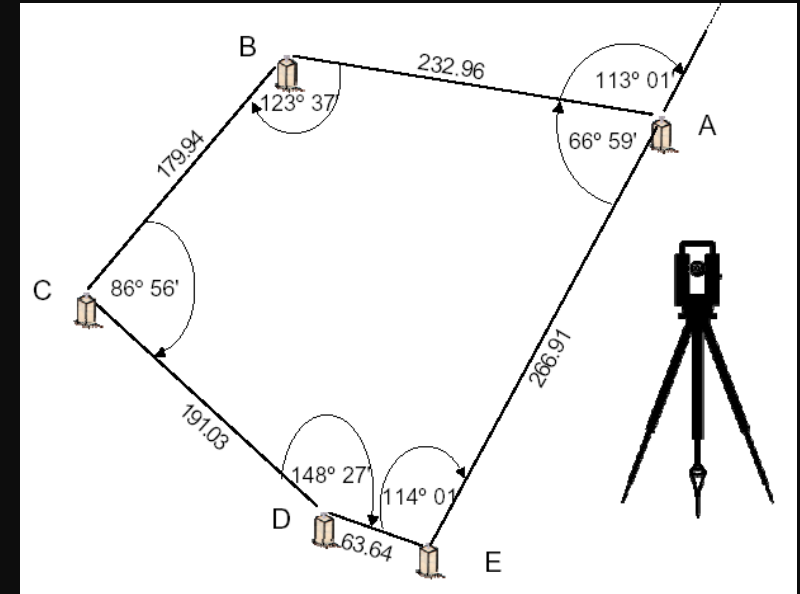


Vector data capture

Primary data collection
(direct measurement)

Ground Surveying

GPS



Theodolite

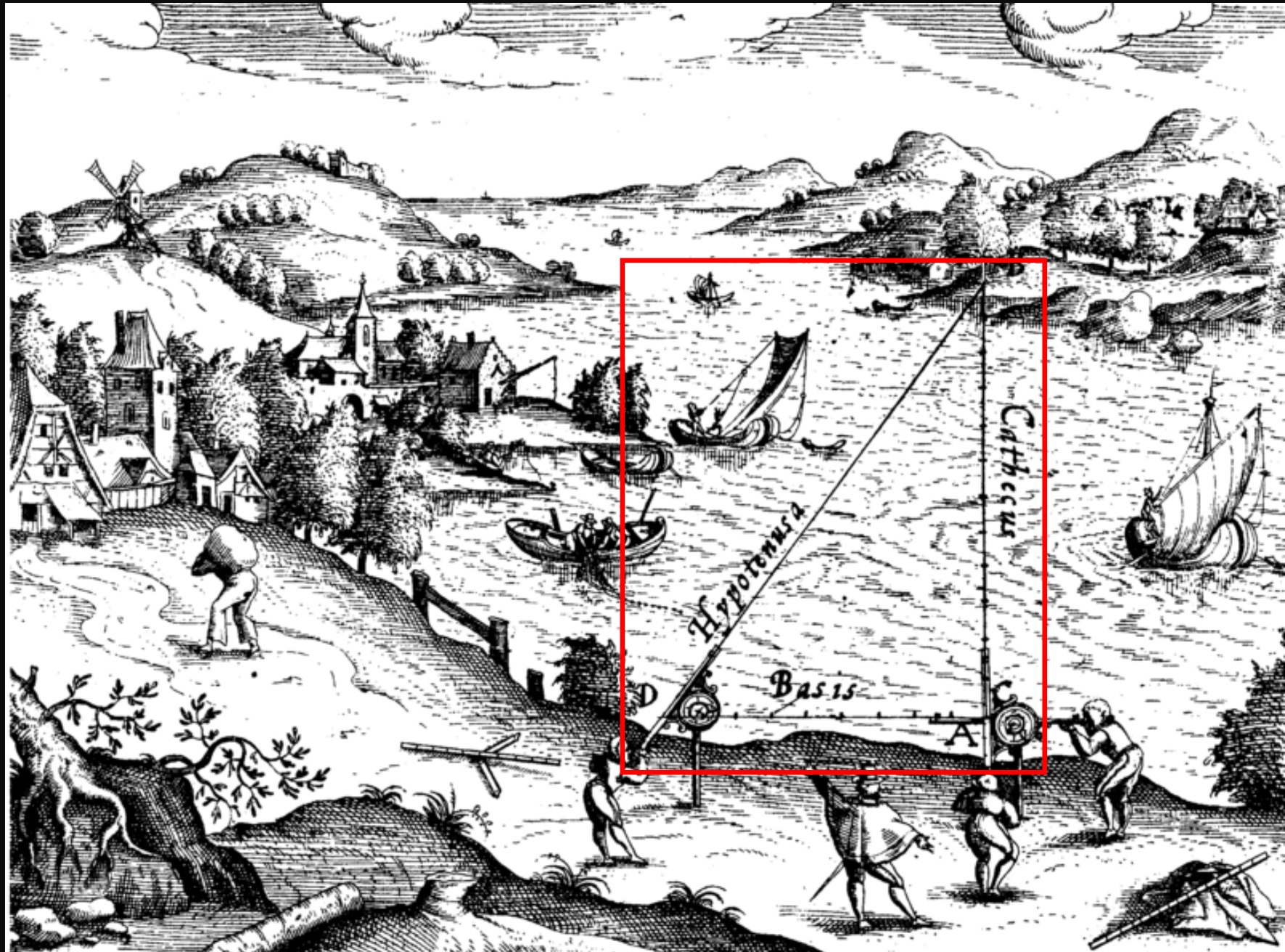


Total Station



Total stations automatically log data and the most sophisticated can create vector point, line and polygon objects in the field, thus providing direct validation.

A survey model



Geo-data from Surveying

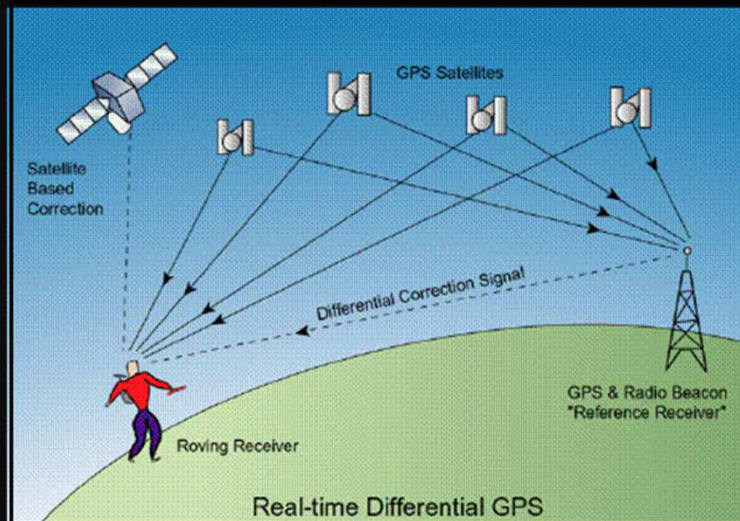
Data collected by field surveying

- Points coordinates
- Distances
- Elevations
- Attribute data

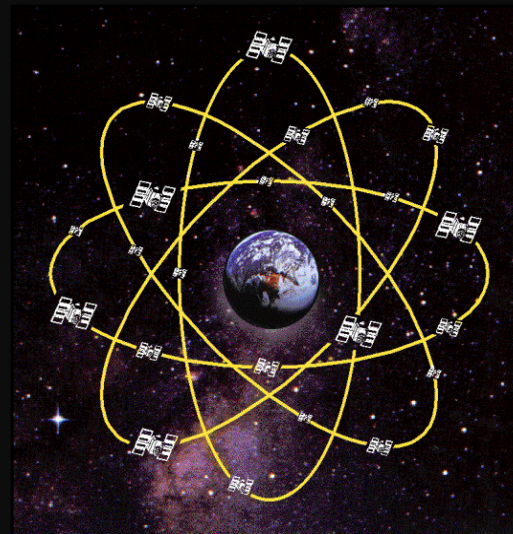
Data entry

- Import, if in digital form
- Manual entry

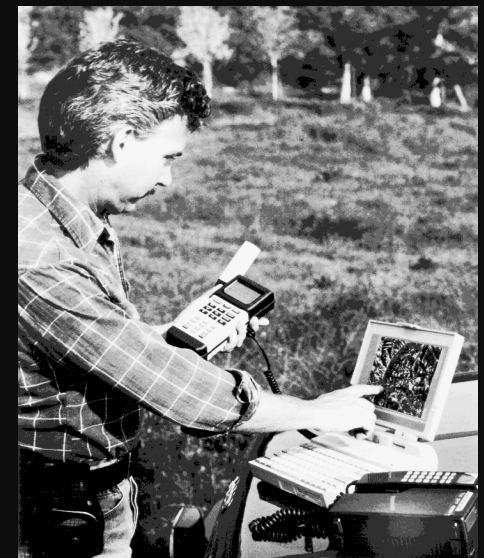
GPS (Global Positioning System)



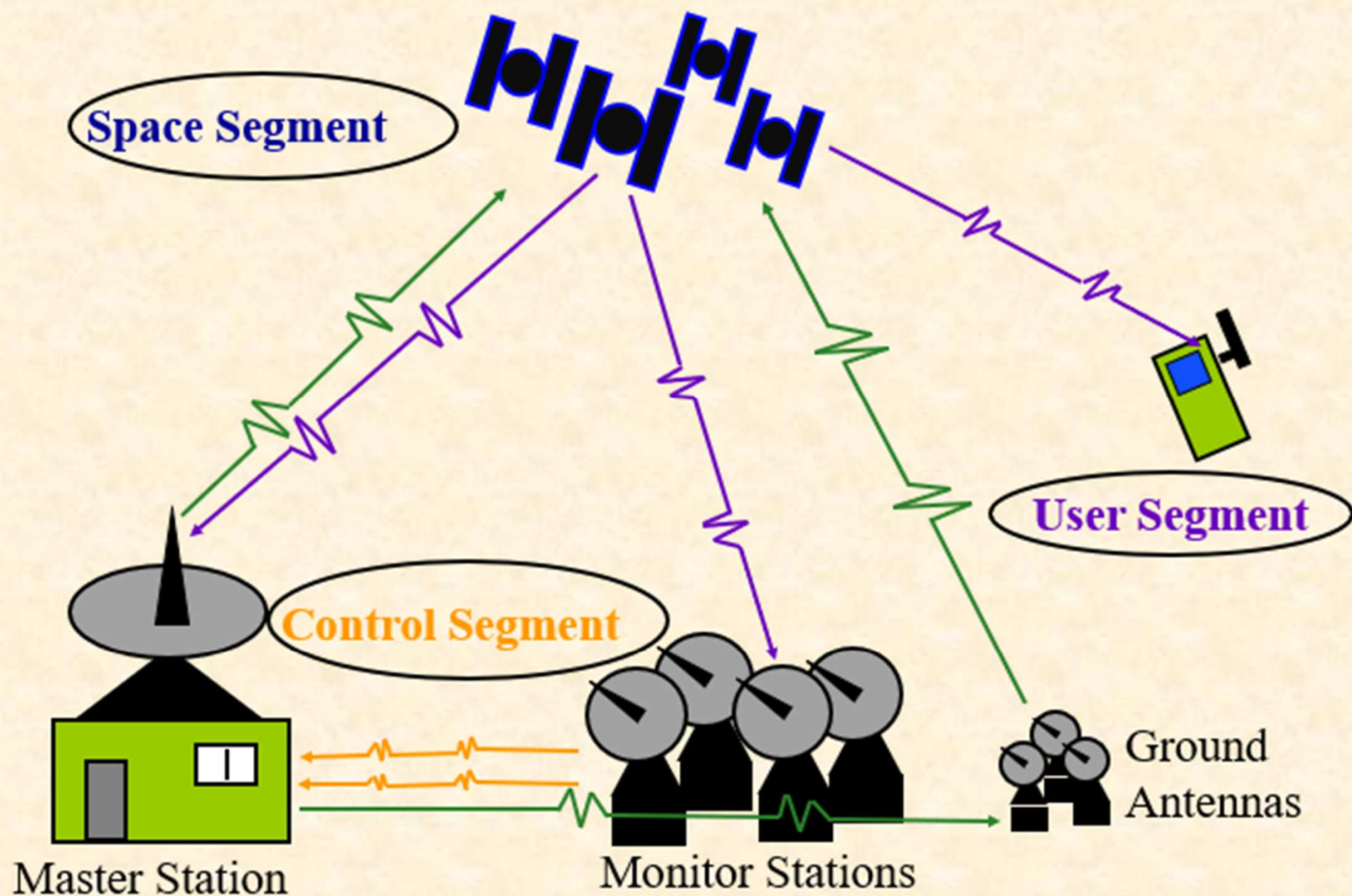
GPS Device



What is GPS?

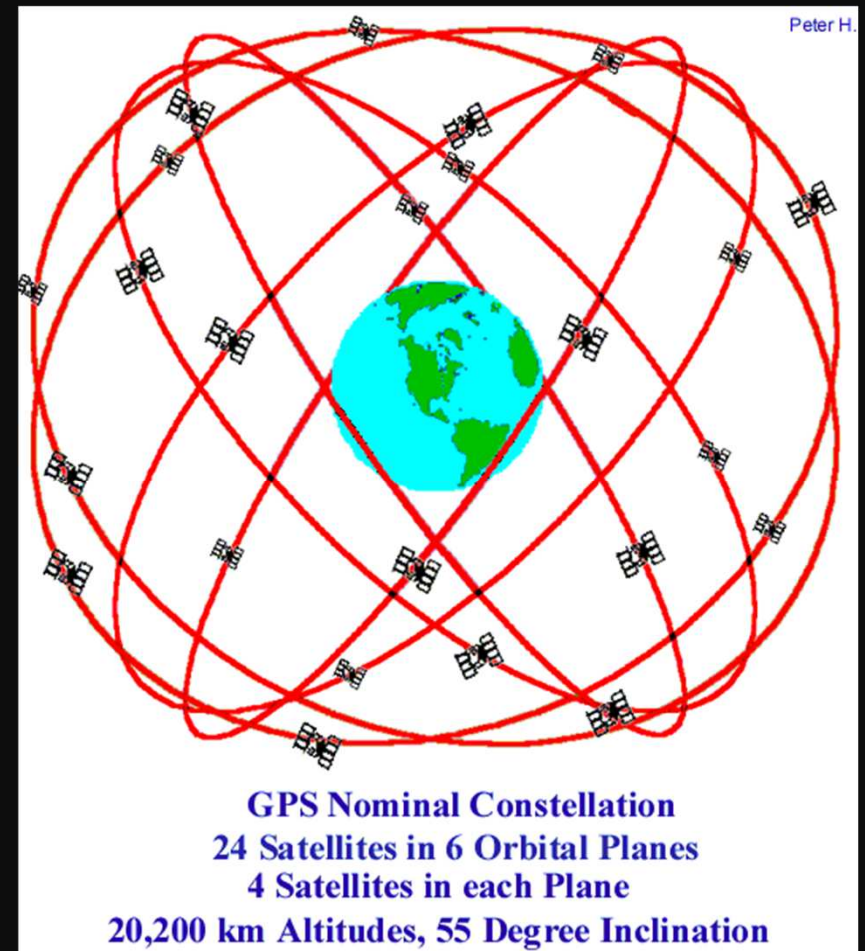
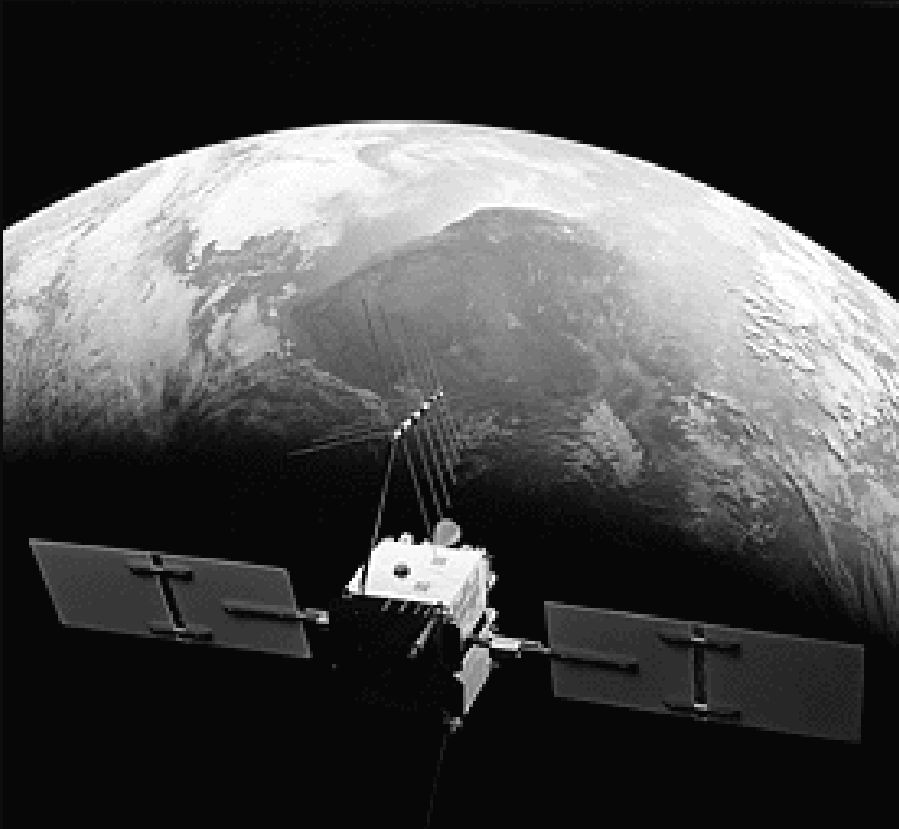


Three Segments of the GPS

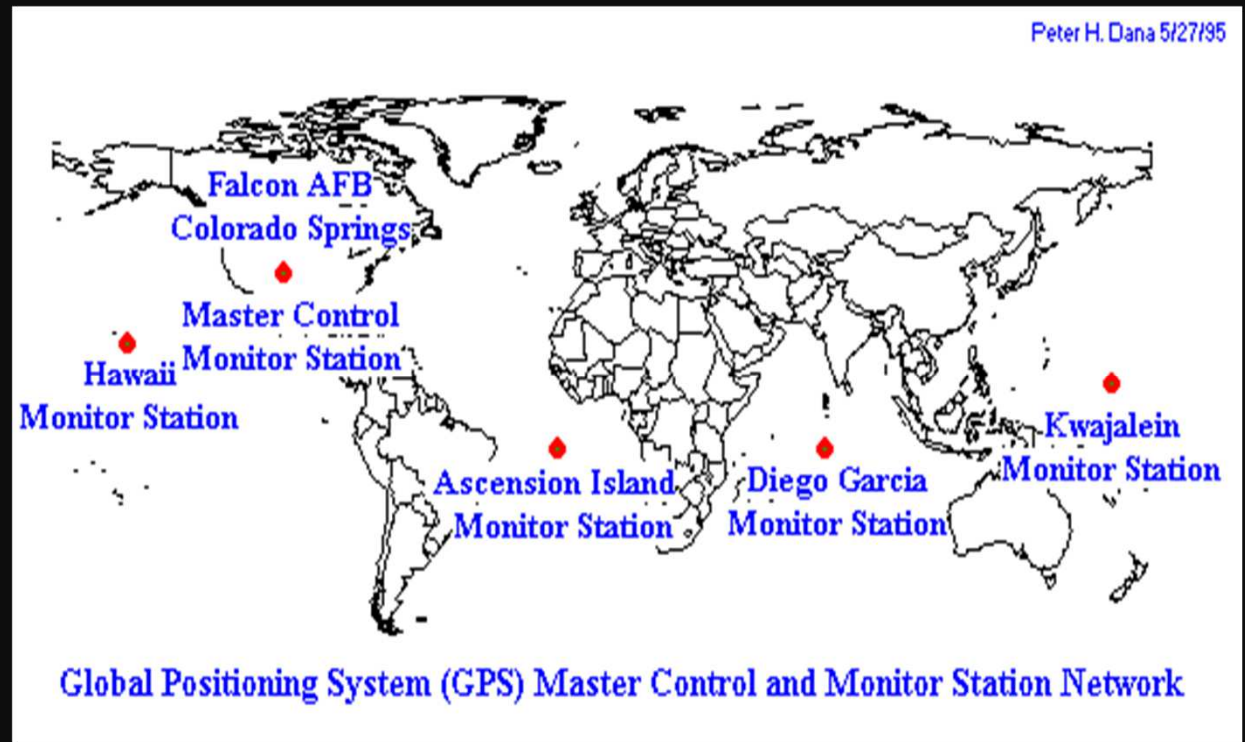


Space Segment

- System consists of 24+ satellites in the operational mode: 21 in use and 3 spares
- Altitude: 20,200 Km with 12 hrs orbit .
- Hydrogen Maser Atomic Clocks



Control Segment

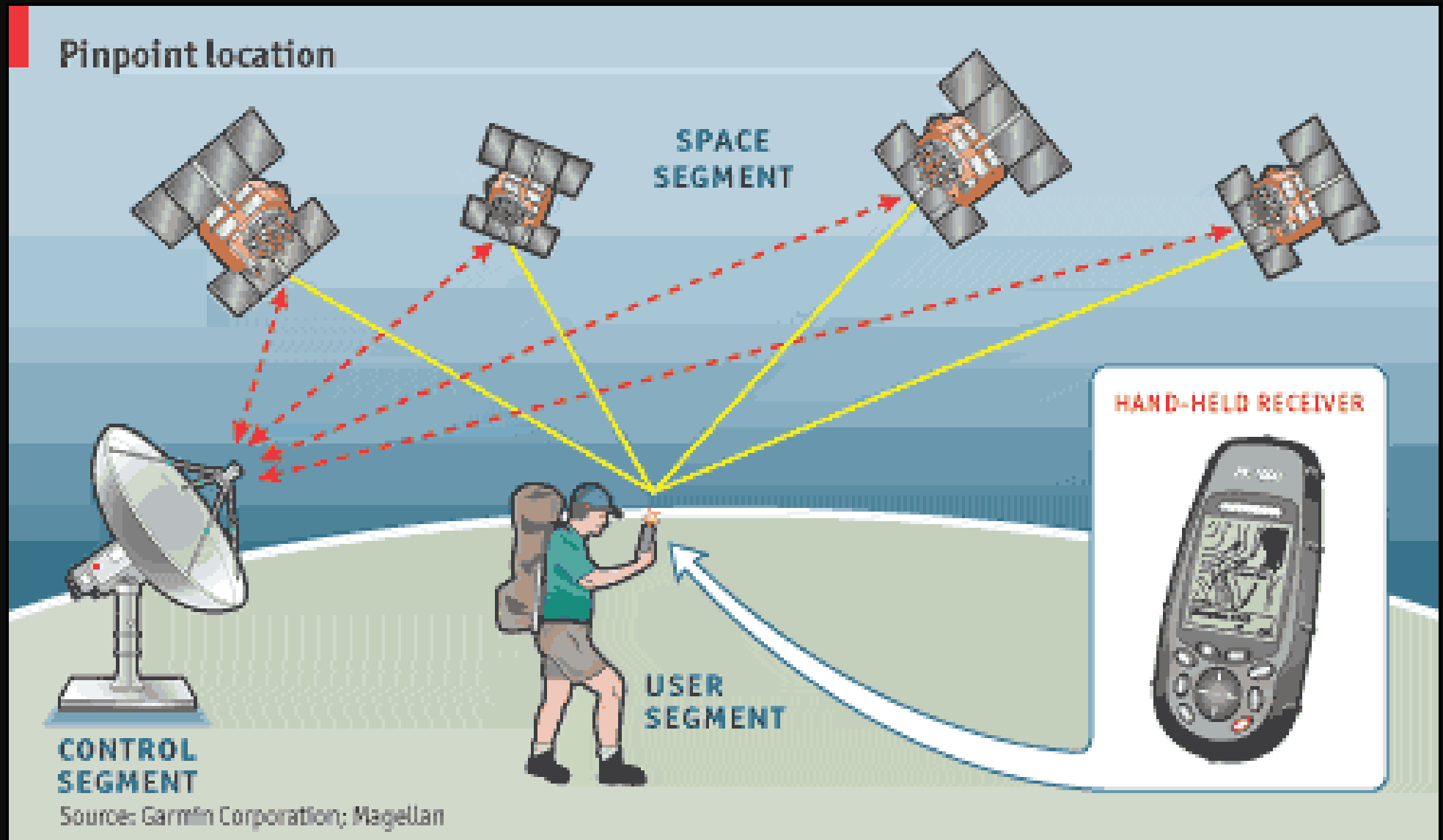


User Segment

- It consists of receivers that decode the signals from the satellites
- The receiver performs following tasks:
 - Selecting one or more satellites
 - Acquiring GPS signals
 - Measuring and tracking
- Used by
 - Ships, Aircraft, Ground vehicles, Individuals



GPS Communication and Control

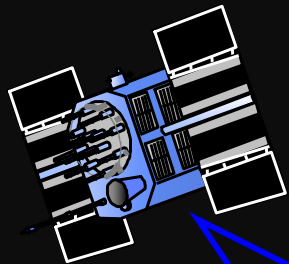


How does GPS work?

$$\text{RATE} \times \text{TIME} = \text{DISTANCE}$$

RATE = Speed of Light (300,000 km/s)

TIME = How long it takes for the signal to reach the receiver

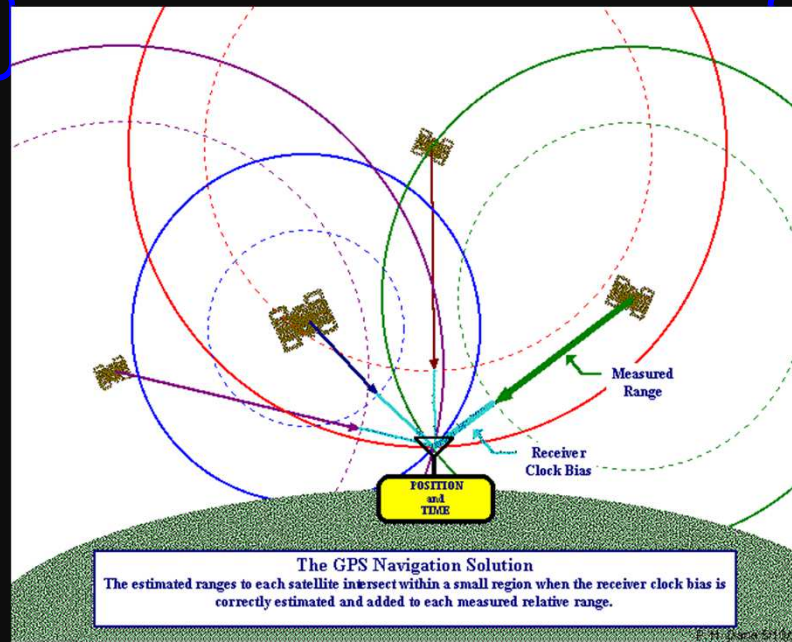


It's 9:30:00

TIME = 1 second
DISTANCE = 300,000 km



Now it's 9:30:01

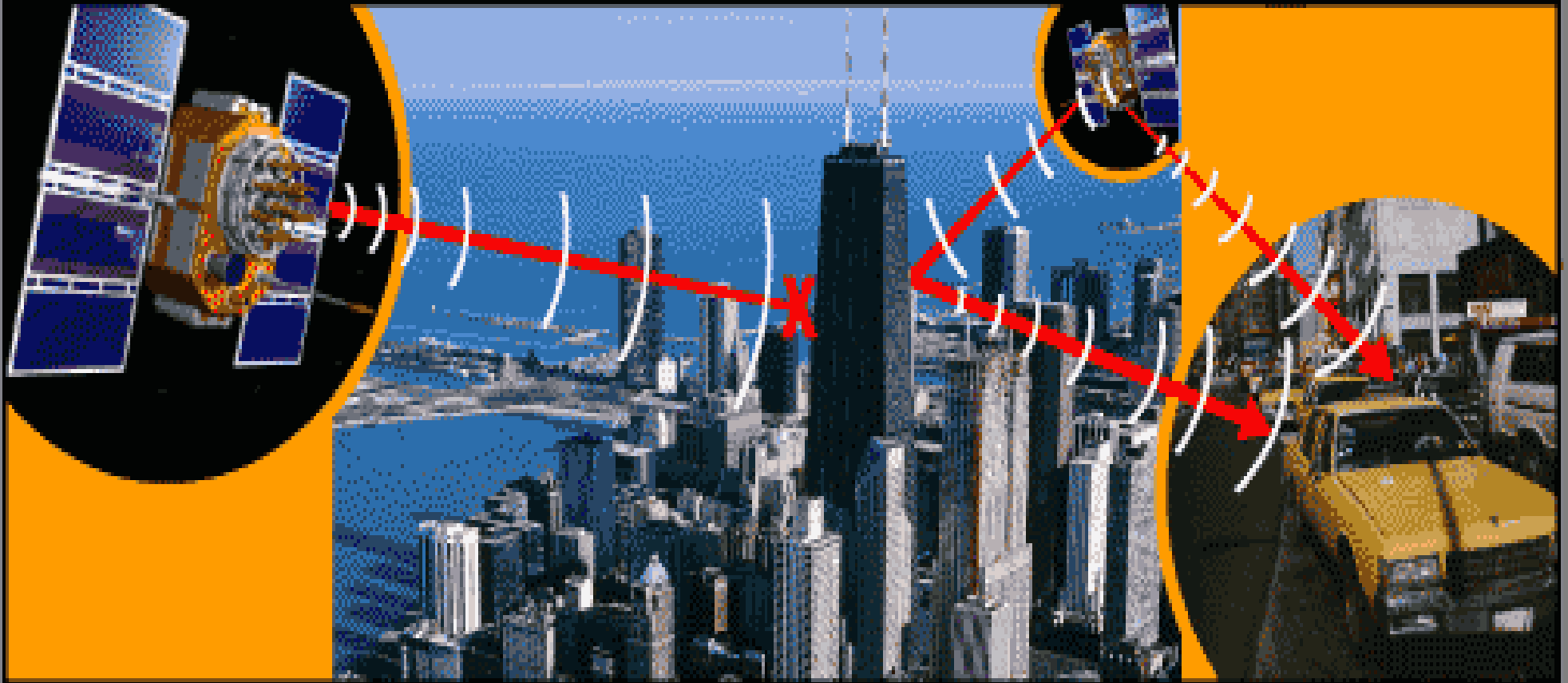


Signal Transmission

1 Blocked Signal

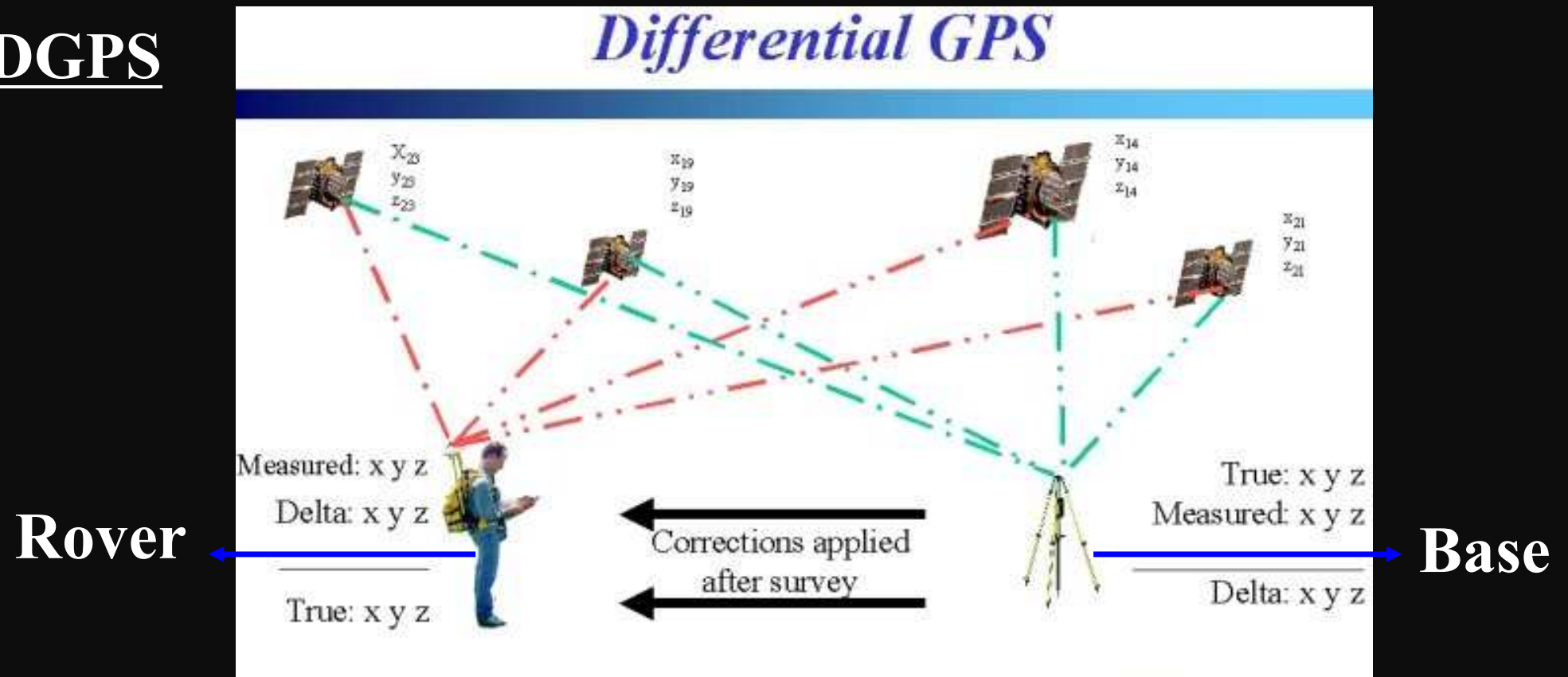
2 Multipath Error

3 Correct Signal



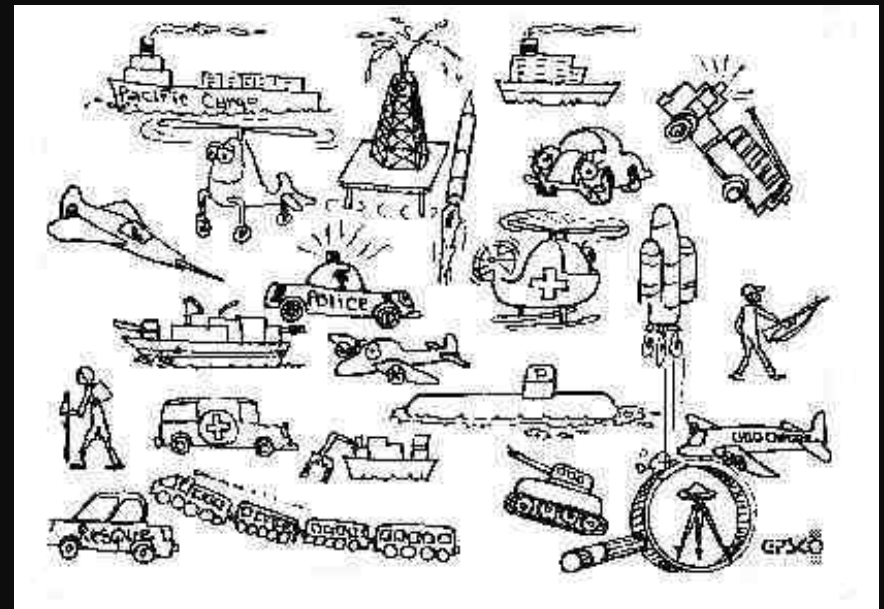
Necessary to have three or more satellites in unobstructed view in order to collect measurements. A problem in **forests** and urban areas with **tall buildings**

DGPS

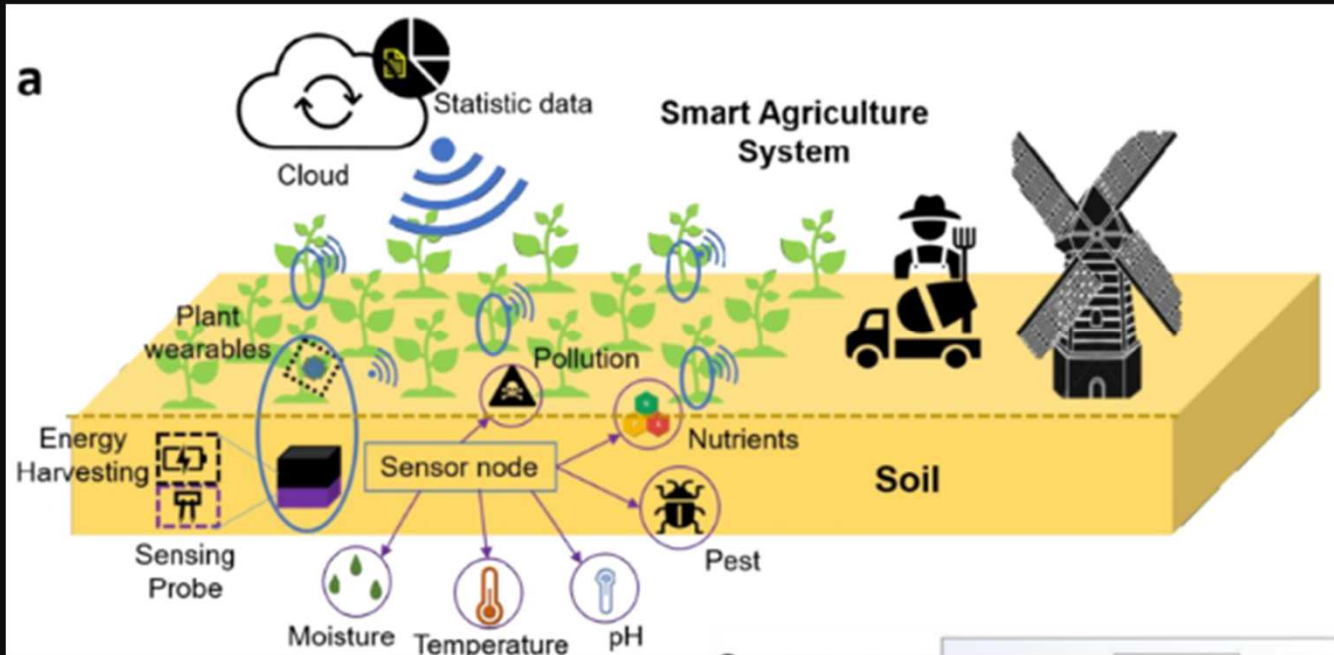


GPS Applications

Navigation, Topographic
surveys, Natural Resource
Management, Other surveys.



Wireless Sensor Networks (WSN)



A wireless sensor network records data such as soil and leaf humidity, as well as air temperature.
Credit: Fraunhofer IMS



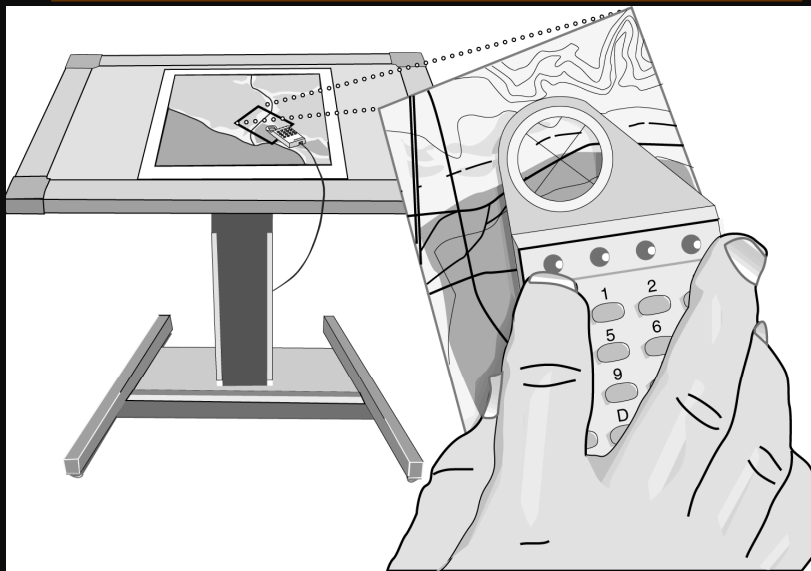
Secondary Vector Data Capture

1. From Raster data

- Scanning of sketches, documents, etc
 - Important scanning parameters are *spatial* and *spectral* (bit depth) resolution
- Digitization
 - The process of converting continuous lines into discrete points so that they can be stored in a computer
 - Manual digitizing / Heads-up digitizing

Tools for Manual digitizing

Digitizing board



1. Digitizer cursor transmits a pulse from an electromagnetic coil under the view lens.
2. Pulse is picked up by nearest grid wires under tablet surface.
3. Result is sent to computer after conversion to x and y units.

Major problems in map digitizing

- The map will stretch or shrink day by day which makes the newly digitized points slightly off from the previous points;
- The map itself has errors;
- Discrepancies across neighboring map sheets will produce disconnectivity;
- Operators will make a lot of errors and mistakes while digitizing.

Heads-up digitizing

Manual - On screen digitization

Steps:

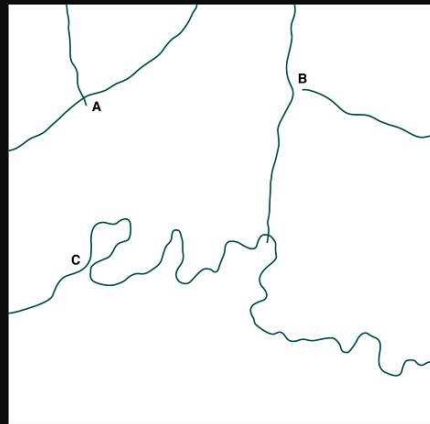
- Source maps or images are scanned ,
- Geo-referenced
- Features traced on screen using mouse and software controls
- Attribute data entered manually.

Typical human errors in digitizing:

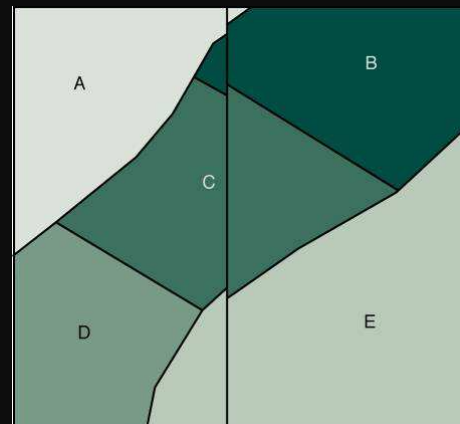
(A) undershoots and overshoots;

(B) invalid polygons;

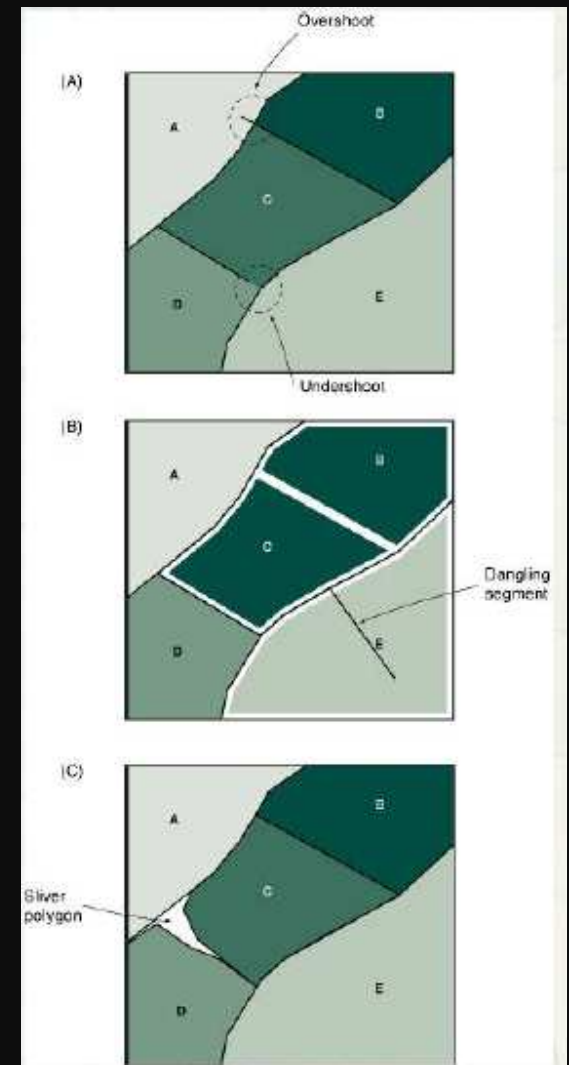
(C) sliver polygons



Error induced by cleaning

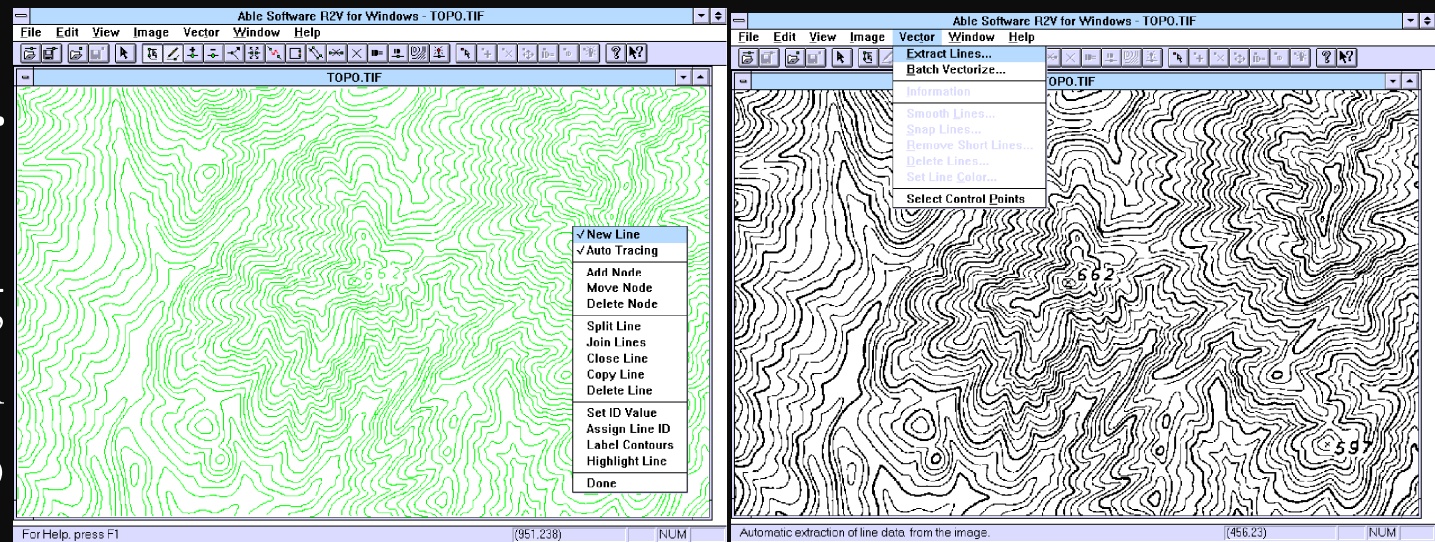


Mismatches



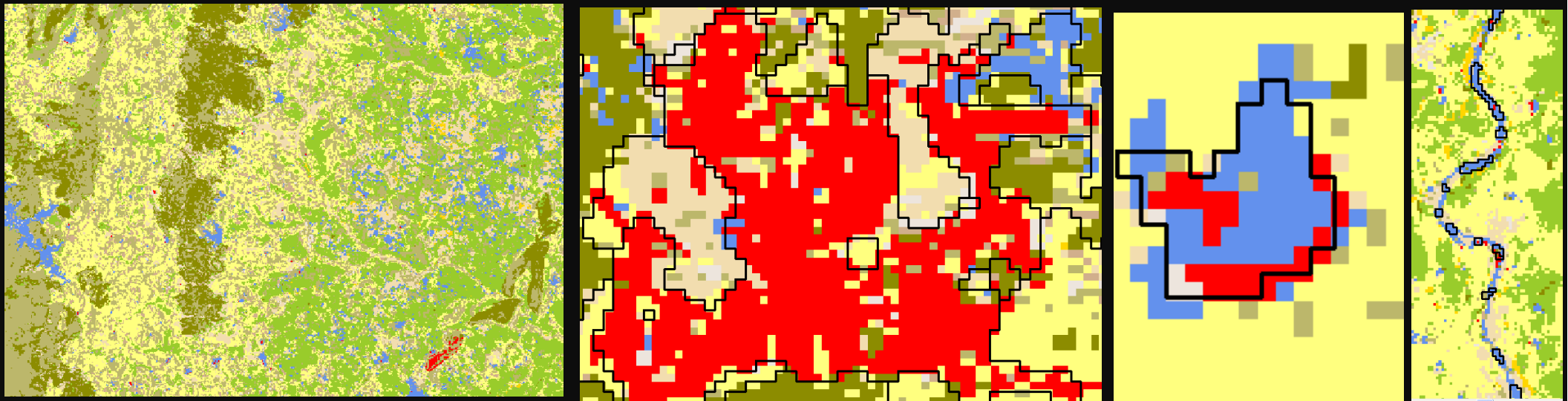
Automated Digitizing Raster Data

Map is scanned using a large format drum scanner converted to binary image to create lines representing features and boundaries.



Software (or user) traces features by following lines: **semiautomatic** or **automatic** (batch mode)

Vectorization of raster files



Text Coordinate Importing

S.No	Long	Lat	Species	Genus	Family	District	Location	Sub-District	Author	
1	93.02558	13.28466	Halimeda c	Halimeda	Chlorophyceae	Andaman	Aerial Bay	North Andaman	Karthick and narayanmurthy2013	
2	93.23235	8.255655	Acanthopora	Acanthopora	Rhodophyta	Nicobar	Bompoka	Nicobar	Ph.d Thesis-Muthuvelan	
3	93.23235	8.255655	Amphiroa	Amphiroa	Rhodophyta	Nicobar	Bompoka	Nicobar	Ph.d Thesis-Muthuvelan	
4	93.23235	8.255655	Caulerpa p	Caulerpa	Chlorophyceae	Nicobar	Bompoka	Nicobar	Ph.d Thesis-Muthuvelan	
5	93.23235	8.255655	Caulerpa r	Caulerpa	Chlorophyceae	Nicobar	Bompoka	Nicobar	Ph.d Thesis-Muthuvelan	
6	93.23235	8.255655	Chaetomor	Chaetomor	Chlorophyceae	Nicobar	Bompoka	Nicobar	Ph.d Thesis-Muthuvelan	
7	93.23235	8.255655	Enteromor	Enteromor	Chlorophyceae	Nicobar	Bompoka	Nicobar	Ph.d Thesis-Muthuvelan	
8	93.23235	8.255655	Gelidiella	Gelidiella	Rhodophyta	Nicobar	Bompoka	Nicobar	Ph.d Thesis-Muthuvelan	
9	93.23235	8.255655	Gracilaria	Gracilaria	Rhodophyta	Nicobar	Bompoka	Nicobar	Ph.d Thesis-Muthuvelan	
10	93.23235	8.255655	Halimeda i	Halimeda	Chlorophyceae	Nicobar	Bompoka	Nicobar	Ph.d Thesis-Muthuvelan	
11	93.23235	8.255655	Halimeda c	Halimeda	Chlorophyceae	Nicobar	Bompoka	Nicobar	Ph.d Thesis-Muthuvelan	
12	93.23235	8.255655	Hypnea m	Hypnea	Rhodophyta	Nicobar	Bompoka	Nicobar	Ph.d Thesis-Muthuvelan	
13	93.23235	8.255655	Laurencia	Laurencia	Rhodophyta	Nicobar	Bompoka	Nicobar	Ph.d Thesis-Muthuvelan	
14	93.23235	8.255655	Lithophyllu	Lithophyllu	Rhodophyta	Nicobar	Bompoka	Nicobar	Ph.d Thesis-Muthuvelan	
15	93.23235	8.255655	Padina gyn	Padina	Pheophyceae	Nicobar	Bompoka	Nicobar	Ph.d Thesis-Muthuvelan	
16	93.23235	8.255655	Padina tet	Padina	Pheophyceae	Nicobar	Bompoka	Nicobar	Ph.d Thesis-Muthuvelan	
17	93.23235	8.255655	Sargassum	Sargassum	Pheophyceae	Nicobar	Bompoka	Nicobar	Ph.d Thesis-Muthuvelan	
18	93.23235	8.255655	Sargassum	Sargassum	Pheophyceae	Nicobar	Bompoka	Nicobar	Ph.d Thesis-Muthuvelan	
19	93.23235	8.255655	Turbinaria	Turbinaria	Pheophyceae	Nicobar	Bompoka	Nicobar	Ph.d Thesis-Muthuvelan	
20	93.23235	8.255655	Turbinaria	Turbinaria	Pheophyceae	Nicobar	Bompoka	Nicobar	Ph.d Thesis-Muthuvelan	
21	93.23235	8.255655	Ulva lactuc	Ulva	Chlorophyceae	Nicobar	Bompoka	Nicobar	Ph.d Thesis-Muthuvelan	
22	92.73276	11.55383	Acanthopora	Acanthopora	Rhodophyta	Andaman	Burmanall	South Andaman	Karthick and narayanmurthy2013	
23	92.73276	11.55383	Acanthopora	Acanthopora	Rhodophyta	Andaman	Burmanall	South Andaman	Karthick and narayanmurthy2013	
24	92.73276	11.55383	Acetabularia	Acetabularia	Chlorophyceae	Andaman	Burmanall	South Andaman	Karthick and narayanmurthy2013	
25	92.73276	11.55383	Actinotricha	Actinotricha	Rhodophyta	Andaman	Burmanall	South Andaman	Karthick and narayanmurthy2013	
26	92.73276	11.55383	Amphiroa	Amphiroa	Rhodophyta	Andaman	Burmanall	South Andaman	Karthick and narayanmurthy2013	
27	92.73276	11.55383	Amphiroa	Amphiroa	Rhodophyta	Andaman	Burmanall	South Andaman	Karthick and narayanmurthy2013	

Participatory / Community-based Spatial Data Collection

VGI (Volunteered Geographic Information): Geographic data voluntarily contributed by individuals, e.g., OpenStreetMap features.

Crowdsourcing: Collecting information from a large number of people, often through apps or online platforms, e.g., citizens reporting road closures or flood locations.

Citizen science: Public participation in scientific data collection, e.g., citizens recording bird observations with eBird or reporting environmental conditions.

Vector Data Transfer

The process of transferring point, line, and polygon data, along with their associated attribute tables, from one GIS/data source to another.

- **File-based transfer** : Shapefile (.shp), GeoPackage (.gpkg), GeoJSON (.geojson), KML/KMZ, and GML
- **Database transfer** : PostGIS, SpatiaLite, Oracle Spatial, and SQL Server Spatial.
- **Web-based transfer** : Vector data can be obtained or transferred through web services such as **WFS (Web Feature Service)**, APIs, and online GIS platforms.
- **Direct software-to-software transfer** : QGIS can import a Shapefile, GeoJSON, KML, or GeoPackage and export it to another format.
- **GNSS/GPS data transfer**: GPX, CSV, or GeoJSON.

Attribute data

Import from GIS databases (ArcGIS, MapInfo etc)

Import from general databases (MS Access, Oracle etc)

Manual entry

Derive new attribute data from existing data
(classification, computation)

Import from field observations

Socio-economic & environmental data

Socio-economic data is widely available, often from national and local government, and is usually the product of population surveys and censuses

This data combines with other datasets to produce neighbourhood Profiles to classify area for marketing purposes -

Geodemographics

Comparison

Attribute	Raster Collection	Vector Collection
Data Nature	Continuous surfaces	Discrete objects
Primary Focus	<i>What</i> is mapping the environment? (Sensors/Grid)	<i>Where</i> are the specific features? (Coordinates)
Storage Penalty	File size scales with grid resolution and area size	File size scales with feature complexity and vertex count
Typical Use	Elevation, temperature, satellite imagery, land cover	Boundaries, roads, pipelines, addresses, customer locations

Field GPS measurement → Primary data capture

Digitizing a paper map → Secondary data capture

Downloading a GeoTIFF → Data transfer

Managing a Data Capture Project

- In any data capture project, there is a fundamental tradeoff between **quality, speed and price**.
- Capturing high-quality data quickly is possible, but it is very expensive. If price is a key consideration, then lower quality data can be captured over a longer period.

